Shock & Cardiogenic Shock: An Approach to a Case of Syncope

Dr. Muhammad Reihan, Internal Medicine | Source: `Reihan_cardiogenic_shock.pdf` (28 slides)

Legend: ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes. The opening-case answer is **reasoned by me; the key is not on the slides**.

🔗 **Bridge from earlier lectures:**

- **Heart Failure (same lecturer):** the **"warm/cold, wet/dry"** profiles; cardiogenic shock is the **cold–wet** extreme of acute HF.
- **CAD: MI is the commonest cause** of cardiogenic shock; **revascularisation**; mechanical complications (**VSD, free-wall rupture, papillary-muscle rupture → acute MR**).
- **Cardiomyopathies:** DCM, myocarditis, **RCM** (listed under obstructive shock here).
- **Pericardial disease: tamponade** (Beck's triad) and constriction = obstructive shock; **pericardiocentesis**.
- **Pulmonary vascular: massive PE** → obstructive shock → **thrombolysis**.
- **Pleural diseases: tension pneumothorax** → needle decompression.
- **Adrenal II: adrenal crisis** = refractory shock → **hydrocortisone**.
- **ARDS lecture:** ECMO; sepsis is the leading cause of both ARDS and distributive shock.

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	⭐ Define shock; **stages** (pre-shock, progressive, end-organ dysfunction)	Stage features
Knowledge	Work-up: routine + aetiology-specific	Lactate; FAST; echo; CTPA; tryptase
Skill	⭐ **Aetiology-directed management** of cardiogenic and obstructive shock	Revascularise MI; pericardiocentesis; thrombolysis; needle decompression
Skill	Stabilisation and **rescue therapy** for refractory shock	Steroids, bicarbonate, MCS/ECMO

Main subtopics:

- overview, definition, pathophysiology, stages
- **types** (cardiogenic, hypovolaemic, obstructive, distributive) and **haemodynamics**
- diagnostics (routine and type-specific)
- **refractory shock** (rescue therapies)

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2. Main Content (Lecture Order)

2.0 Opening Case (Slides 5–6)

Presentation:

- **28-year-old** having a skin graft; tibial ORIF 4 weeks ago
- **spinal anaesthesia: intrathecal bupivacaine at L3–L4**
- **30 minutes later:** dyspnoea, light-headedness, **loss of consciousness**
- **HR 38 (bradycardia)**, **BP 82/50**, RR 24, **flushed skin**
- withdraws the **upper limbs only to pain above the navel** (sensory level ~T10 or higher)
- **lungs clear**

Most likely cause? → ⭐ **A. Loss of sympathetic vascular tone** *[reasoning added; key not on slides]*:

- a **high spinal block** blocks sympathetic outflow (T1–L2) → **vasodilatation (↓ SVR), venous pooling**
- if it reaches **T1–T4 (cardioaccelerator fibres)** → **bradycardia**
- this is **neurogenic/distributive shock** physiology: **hypotension + bradycardia + warm, flushed skin**
- *[added]* treat with fluids, **vasopressors (phenylephrine/ephedrine)**, **atropine**, airway support

Why not the others:

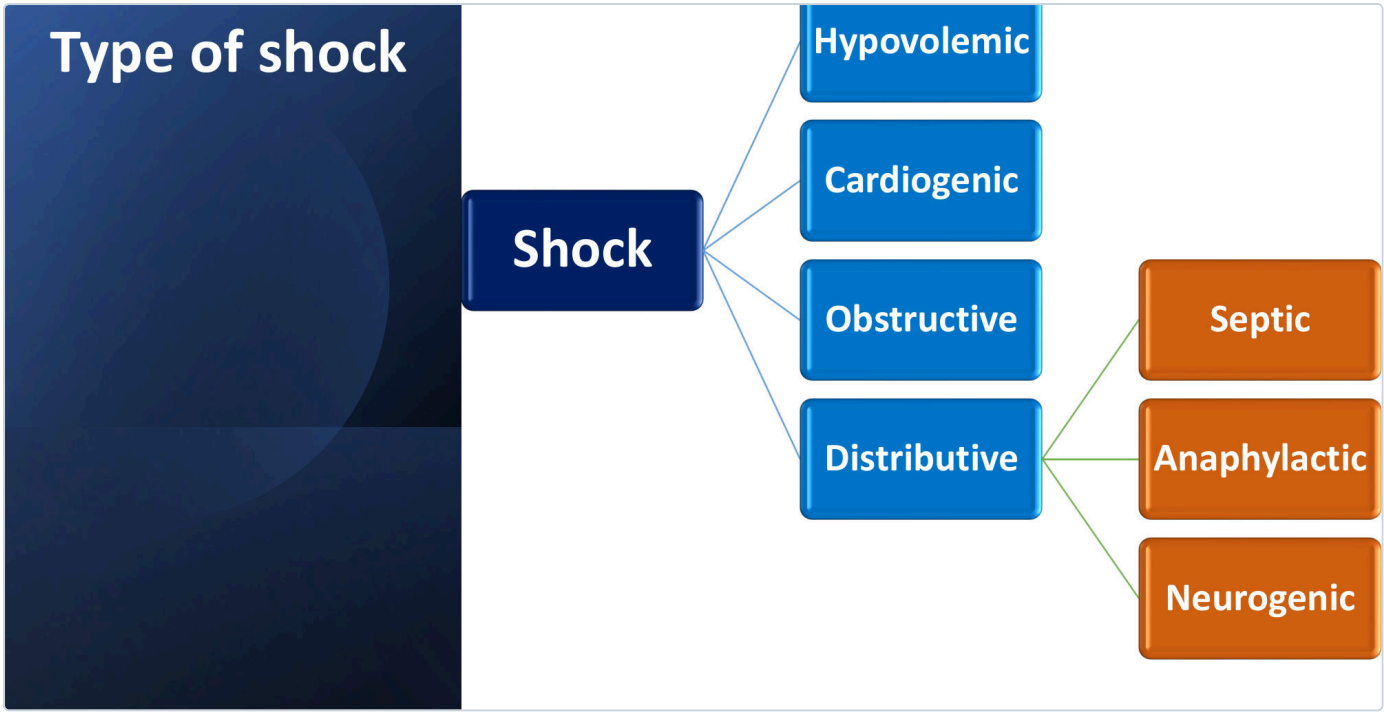
- **B (septic)** and **F (anaphylaxis)** cause **tachycardia** (anaphylaxis also has urticaria and wheeze)
- **C (PE)** → tachycardia, hypoxaemia, raised JVP
- **D (hypovolaemia)** → **cold** skin with tachycardia
- **E (fat/marrow embolism)** → within **24–72 h** of a fracture, with petechiae, hypoxaemia and confusion; the fracture was 4 weeks ago

2.1 Definition and Types (Slides 7–8) ⭐

⭐ **Shock** = a **life-threatening circulatory disorder** → **tissue hypoxia** and **microcirculatory disturbance** → **metabolic derangement** → ultimately **irreversible organ damage**.

[added] In simple terms: **O_2 delivery < demand**. **MAP = CO × SVR.**

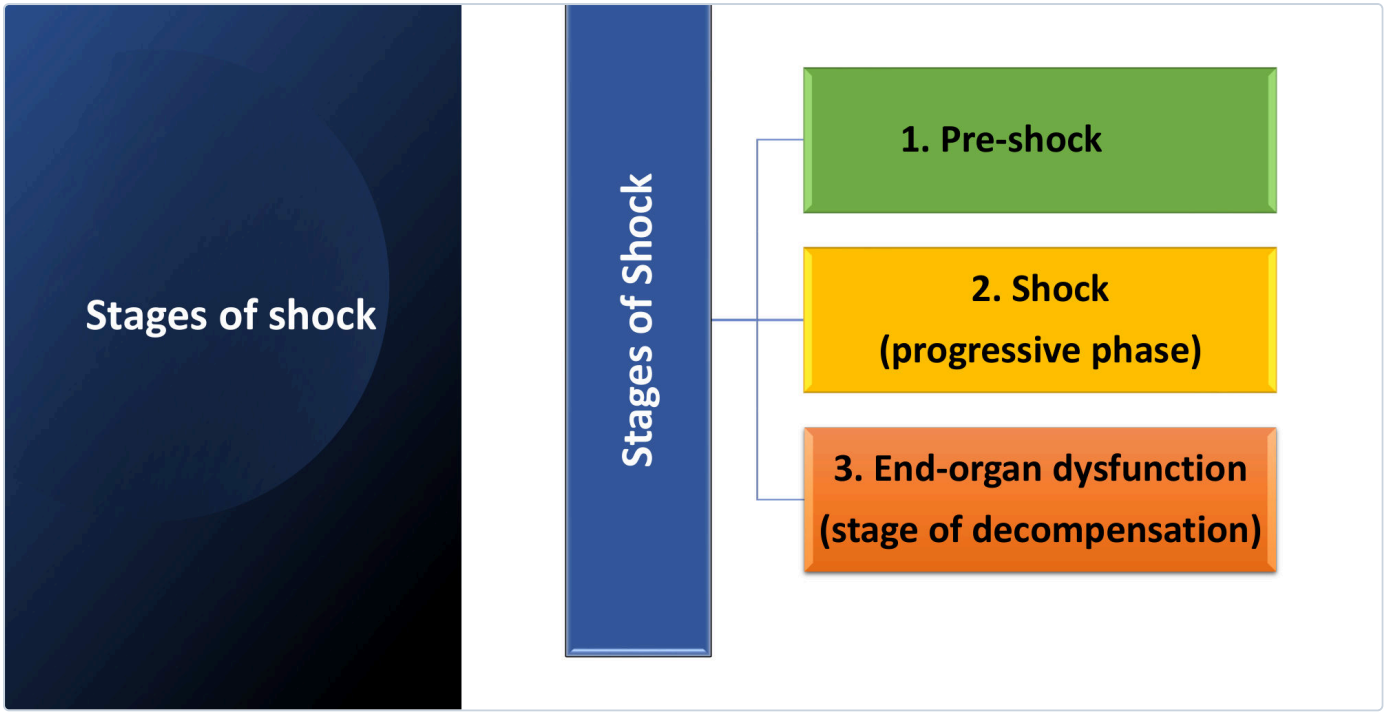
Type	Subtypes / examples
Hypovolaemic	Haemorrhagic; non-haemorrhagic (GI, renal, third-space, insensible losses)
⭐ **Cardiogenic**	Pump failure (MI…)
Obstructive	Tamponade, tension PTX, massive PE…
Distributive	⭐ **Septic · anaphylactic · neurogenic**



Slide 8: Types of shock

2.2 Stages of Shock (Slides 9–12) ⭐

Stage	Features
1. Pre-shock (compensated)	⭐ **Peripheral vasoconstriction**: **cold, clammy extremities, ↑ capillary refill time** (*may be absent in distributive shock*). ↓ capillary hydrostatic pressure → **interstitial fluid drawn into the vessels** (autotransfusion). ⭐ **Tachycardia** (*neurogenic shock may be bradycardic*). **Oliguria**
2. Shock (progressive)	**Worsening hypotension**; tissue hypoperfusion → **worsening tachypnoea**. **Precapillary dilation + postcapillary constriction** → capillary pooling and stasis → ↓ CO + **microthrombi → DIC**. **Acidosis**, cerebral hypoperfusion → **altered mental status**
3. End-organ dysfunction (decompensation)	Cerebral hypoxia → **autonomic dysfunction**; **myocardial ischaemia → ↓ CO**; widespread necrosis → **lysosomal enzymes** (more injury) and **cytokines** (DIC); **bowel ischaemia → bacteraemic sepsis**; ⭐ a **vicious cycle → irreversible multi-organ failure**



Slide 9: Stages of shock

2.3 Haemodynamic Parameters (Slide 13) ⭐⭐

Type	CO	Preload (PCWP)	Afterload (SVR)
Hypovolaemic	↓	⭐ ↓	↑
⭐ **Cardiogenic**	↓↓	⭐ ↑	↑
Obstructive	↓↓	**↓ or ↑** (*[added]* ↑ RA/CVP; PCWP ↓ in PE, equalised ↑ in tamponade)	↑
Distributive	**Neurogenic ↓**; **septic/anaphylactic: early ↑, late ↓**	**Usually ↓**	⭐ ↓↓

Key patterns:

- ⭐ **Cardiogenic = the only one with high PCWP** (wet lungs).
- ⭐ **Distributive = the only one with low SVR** (warm peripheries).

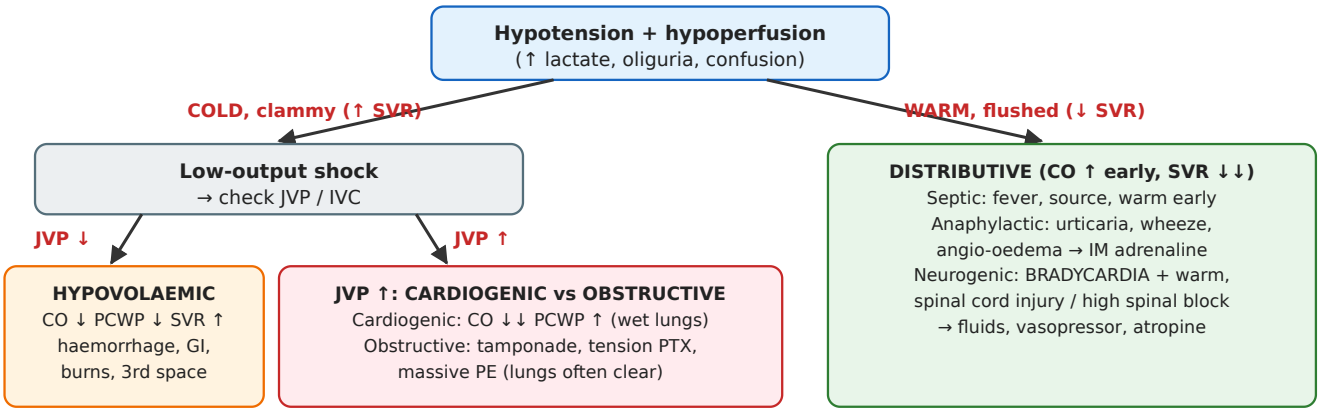
[added] **Mixed venous O_2 (SvO_2):** ↓ in low-output shock (hypovolaemic, cardiogenic, obstructive); ↑ in septic shock (poor O_2 extraction).

Typical hemodynamic parameters of types of shock [3][4][5][6]

Type	Estimated cardiac output (CO)	Estimated preload (e.g., PCWP)	Estimated afterload (e.g., SVR)
Hypovolemic	• ↓	• ↓	• ↑
Cardiogenic	• ↓↓	• ↑	• ↑
Obstructive	• ↓↓	• ↓ Or ↑	• ↑
Distributive	• Neurogenic: ↓ • Septic and anaphylactic ◦ Early: ↑ ◦ Late: ↓	• Usually ↓	• ↓↓

Slide 13: Haemodynamic parameters by shock type

Shock at the bedside: skin temperature + JVP/IVC sort the four types



Cardiogenic (MI commonest): revascularise · norepinephrine → + dobutamine/milrinone if hypoperfused · diuretic once SBP > 90

Cold-dry: small fluid challenge (250–500 mL) if PCWP < 15 → vasopressor → inotrope

Cold-wet: inotrope ± norepinephrine; NO fluids; MCS (IABP/Impella/VA-ECMO) if refractory

Obstructive: pericardiocentesis · needle decompression → chest drain · thrombolysis for massive PE

Hypovolaemic: stop bleeding, blood products (1:1:1), crystalloid for non-haemorrhagic loss

Refractory shock: hydrocortisone · bicarbonate (severe acidosis) · mechanical circulatory support / ECMO

MAP target ≥ 65 mmHg; lactate trend; urine output ≥ 0.5 mL/kg/h [targets added]

Added diagram: Bedside classification and treatment of shock

🔁 **Recap: Shock basics.**

- Shock = tissue hypoxia from circulatory failure.
- Four types (distributive = septic, anaphylactic, neurogenic).
- Stages: compensated (cold, tachycardic, oliguric) → progressive (hypotension, acidosis, DIC, confusion) → end-organ failure (vicious cycle).
- **Cardiogenic: CO ↓↓, PCWP ↑, SVR ↑. Distributive: SVR ↓↓.**

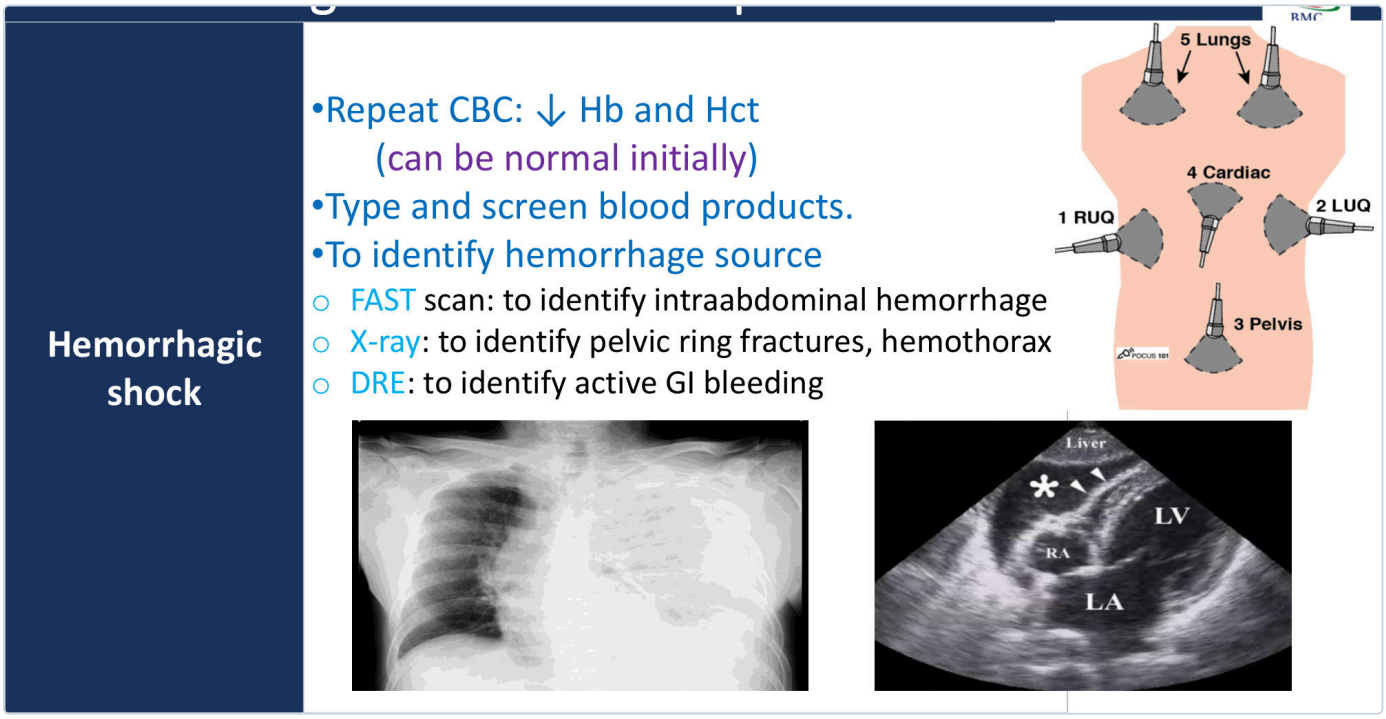
▶️ **Up next: Diagnostics.**

2.4 Diagnostics (Slides 14–19) ⭐

Routine:

- **global hypoperfusion:** ⭐ **↑ lactate**, **ABG**
- **aetiology: BMP**
- **end-organ damage: renal function, LFTs, coagulation panel**

Shock type	Specific studies
⭐ **Haemorrhagic**	**Repeat CBC** (↓ Hb/Hct, *can be normal initially*); **type and screen**. Find the source: ⭐ **FAST** (intra-abdominal blood; *[added]* RUQ, LUQ, pelvis, cardiac, lungs), **X-ray** (pelvic ring fracture, haemothorax), **DRE** (GI bleeding)
Non-haemorrhagic hypovolaemic	**Insensible losses:** fever work-up. **Third-space: CT abdomen** (pancreatitis, bowel obstruction). **GI:** nausea, vomiting, diarrhoea. **Renal:** urine electrolytes, urine creatinine
⭐ **Cardiogenic**	**BNP/NT-proBNP vs baseline**; **ECG** (MI, arrhythmia, ventricular hypertrophy); **troponin I/T** (ACS); ⭐ **echo**: valvular lesions, ventricular dysfunction, **pericardial effusion**
⭐ **Obstructive**	**Echo/POCUS** (depends on cause); **CXR or CT chest** (pneumothorax); ⭐ **CTPA or V/Q** (PE), CT for aortic dissection
Septic	**Cultures** (blood, urine); **ESR, CRP, procalcitonin**
Anaphylactic	Mostly **clinical**; ⭐ **serum mast-cell tryptase** if uncertain
Neurogenic	**CT/MRI spine or brain** to find the injury



Slide 15: Haemorrhagic shock work-up (FAST)

2.5 Cardiogenic Shock: Aetiology and Pathophysiology (Slides 20–21) ⭐⭐

Causes:

- ⭐ **myocardial infarction: the most common cause** (*[added]* usually ≥ 40% of LV lost, or a mechanical complication)
- **arrhythmias**

- **heart failure** (decompensated)
- **cardiomyopathy**
- **myocarditis**
- ⭐ **VSD, ventricular (free-wall) rupture** (*[added]* post-MI days 3–5)
- **valve defects: severe AR or MR** (*[added]* papillary-muscle rupture, IE)
- **blunt cardiac trauma**
- ⭐ **drugs: β-blockers, calcium-channel blockers** (overdose)

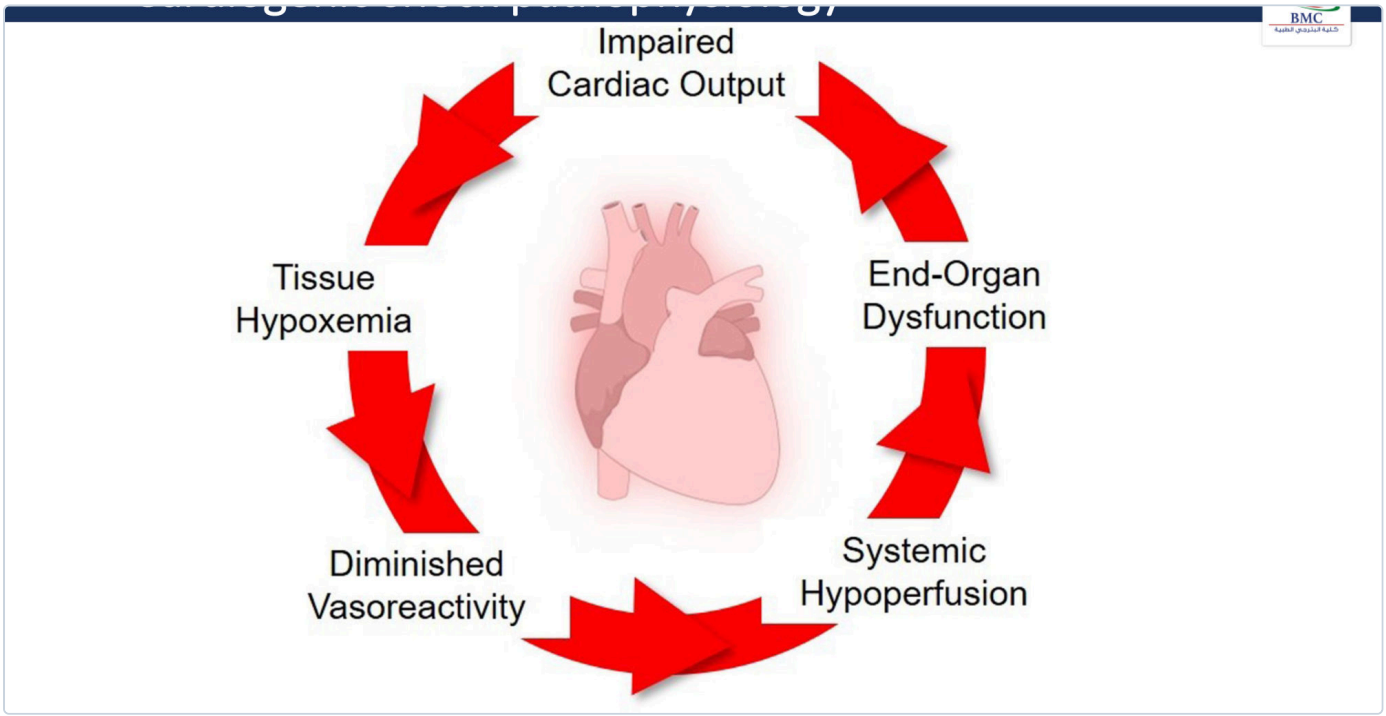
Mnemonic *[added]*: "MI PUMPS":

- **MI** (± mechanical complications)
- **P**apillary/valve rupture (acute MR/AR)
- **U**nstable rhythm (arrhythmia)
- **M**yocarditis / cardio**M**yopathy
- **P**oisoning (β-blocker/CCB)
- **S**eptal rupture (VSD) / free-wall rupture

Downward spiral (slide 21):

1. **impaired CO**
2. **systemic hypoperfusion**
3. **diminished vasoreactivity**
4. **tissue hypoxaemia**
5. **end-organ dysfunction**
6. → further ↓ CO

[added] Coronary perfusion falls with hypotension → more ischaemia → a worse pump.



Slide 21: The vicious cycle of cardiogenic shock

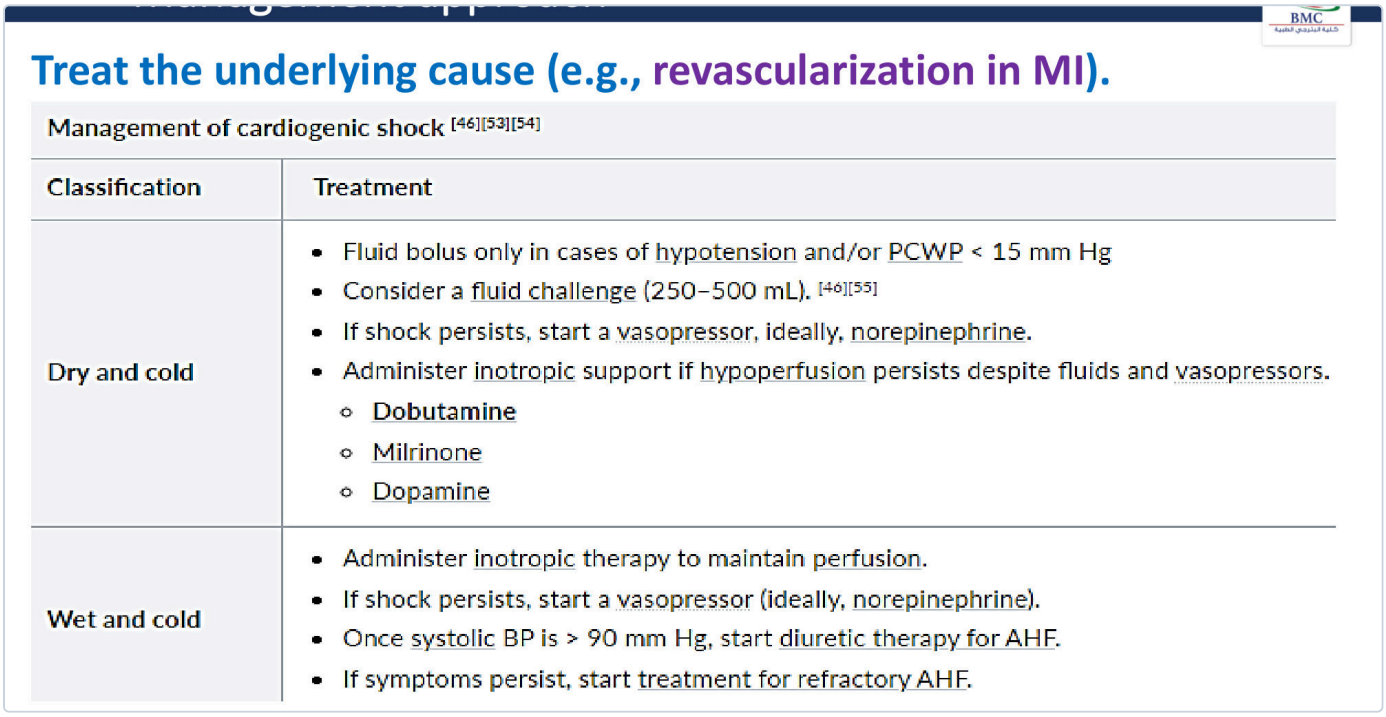
2.6 Cardiogenic Shock: Management (Slide 22) ⭐⭐

⭐ **Treat the underlying cause** (e.g. **revascularisation in MI**: *[added]* emergency PCI of the culprit vessel, SHOCK trial).

Profile	Treatment
⭐ **Dry and cold** (low CO, low filling pressure)	**Fluid bolus only if hypotensive and/or PCWP < 15 mmHg**; consider a **fluid challenge (250–500 mL)** → if shock persists, a **vasopressor, ideally norepinephrine** → if hypoperfusion persists despite fluids and pressors, **inotropes**: **dobutamine**, **milrinone**, **dopamine**
⭐ **Wet and cold** (classic cardiogenic shock)	**Inotropic therapy** to maintain perfusion → if shock persists, a **vasopressor (norepinephrine)** → ⭐ **once SBP > 90 mmHg, start diuretics** for acute HF → if symptoms persist, **refractory AHF** treatment

[added]

- **Norepinephrine is the first-line pressor**: dopamine causes more arrhythmias (SOAP II), which is why the slide lists dopamine only as an inotrope option.
- **Avoid: β-blockers, nitrates/vasodilators, negative inotropes** while in shock.
- **Mechanical circulatory support** if refractory: **IABP** (no mortality benefit in IABP-SHOCK II), **Impella**, **VA-ECMO**.
- **Targets: MAP ≥ 65**, falling lactate, urine output ≥ 0.5 mL/kg/h.



Slide 22: Management of cardiogenic shock (dry–cold vs wet–cold)

2.7 Obstructive Shock (Slides 23–24) ⭐⭐

Mechanism	Causes
↓ Diastolic filling	⭐ **Cardiac tamponade**, **constrictive pericarditis**, **restrictive cardiomyopathy**
↓ Venous return	⭐ **Tension pneumothorax**, intrathoracic tumour
↑ Ventricular afterload	⭐ **Massive PE**, **aortic dissection**, **aortic stenosis**, large systemic emboli, **severe pulmonary hypertension**

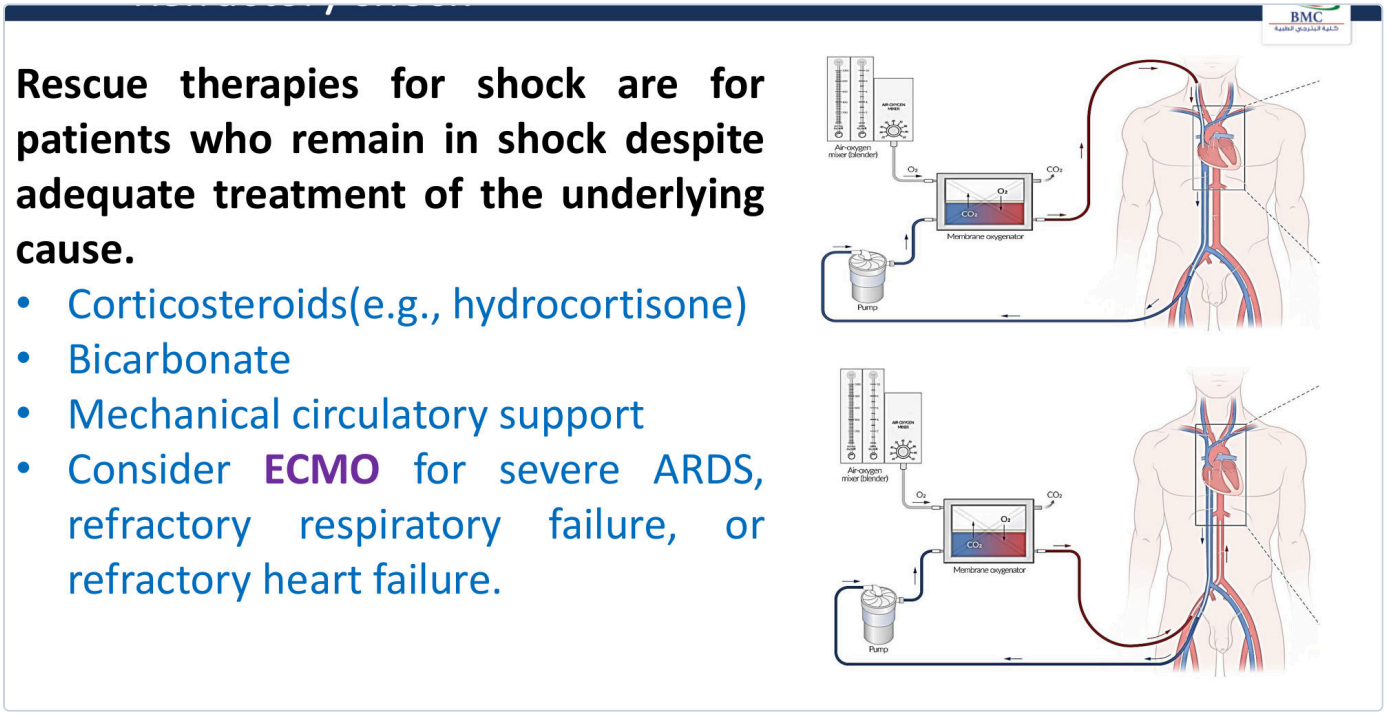
Cause	Treatment (slide 24)
⭐ **Cardiac tamponade**	**Pericardiocentesis**
⭐ **Massive PE**	**Thrombolytic therapy** (*[added]* or embolectomy if contraindicated)
⭐ **Tension pneumothorax**	**Needle thoracostomy → chest tube** (or surgery)

[added] **Bedside clues:**

- **Tamponade: Beck's triad** (hypotension, ↑ JVP, muffled heart sounds), **pulsus paradoxus**, electrical alternans.
- **Tension PTX:** absent breath sounds, hyperresonance, **tracheal deviation away**, ↑ JVP. A **clinical** diagnosis: decompress before the X-ray.
- **Massive PE:** ↑ JVP, RV dilatation on POCUS (McConnell's sign), hypoxaemia, S1Q3T3.

2.8 Complications and Refractory Shock (Slides 25–26) ⭐

- **Complications:**
  - **general: cardiovascular collapse**, **acute renal failure**
  - **specific: pulmonary oedema** (cardiogenic)
  - *[added]* also ARDS, DIC, ischaemic hepatitis ("shock liver"), mesenteric ischaemia
- ⭐ **Refractory shock** = still in shock **despite adequate treatment of the underlying cause** → **rescue therapies:**
  1. ⭐ **corticosteroids (hydrocortisone)** (*[added]* septic shock on escalating pressors; adrenal insufficiency)
  2. **bicarbonate** (*[added]* severe metabolic acidosis, pH < 7.2 with AKI)
  3. ⭐ **mechanical circulatory support**
  4. ⭐ **ECMO** for **severe ARDS, refractory respiratory failure, or refractory heart failure** (*[added]* **VV-ECMO** = lung support; **VA-ECMO** = heart + lung support)



Slide 26: Refractory shock (ECMO circuits)

🔁 **Recap: Cardiogenic and obstructive shock.**

- **Cardiogenic:** MI commonest (plus arrhythmia, HF, CM, myocarditis, VSD/rupture, AR/MR, trauma, β-blocker/CCB). Echo, troponin, ECG, BNP.
- **Revascularise.**
- Dry–cold: small fluids → norepinephrine → inotrope.
- Wet–cold: inotrope → norepinephrine → diuretic when SBP > 90.
- **Obstructive:** tamponade (pericardiocentesis), tension PTX (needle → drain), massive PE (thrombolysis).
- **Refractory:** hydrocortisone, bicarbonate, MCS/ECMO.

3. High-Yield Comparison Tables

Table A: The Four Shock Types

	Hypovolaemic	Cardiogenic	Obstructive	Distributive
CO	↓	**↓↓**	↓↓	↑ early (septic/anaphylaxis), ↓ neurogenic/late
PCWP	**↓**	**↑**	↓ or ↑	↓
SVR	↑	↑	↑	**↓↓**
JVP	Flat	**↑**	**↑**	Flat/normal
Skin	Cold	Cold	Cold	**Warm**
Lungs	Clear	**Crackles**	Clear (PE, tamponade) / absent sounds (PTX)	Clear (± wheeze in anaphylaxis)
First treatment	**Volume/blood**	**Revascularise; inotrope ± norepinephrine**	**Relieve the obstruction**	**Fluids + norepinephrine** (septic); **IM adrenaline** (anaphylaxis); fluids + pressor + atropine (neurogenic)

Table B: Distributive Shock Subtypes *[slides 8, 10, 19 + added]*

	Septic	Anaphylactic	Neurogenic
Trigger	Infection	Allergen (IgE)	**Spinal cord injury (≥ T6)**, **high spinal/epidural block**
HR	**Tachycardia**	**Tachycardia**	⭐ **Bradycardia**
Skin	Warm (early) → cold (late)	Urticaria, flushing, angio-oedema	**Warm, dry, flushed**
Test	Cultures, lactate, procalcitonin	**Tryptase**	CT/MRI spine
Treatment	Antibiotics within 1 h, 30 mL/kg crystalloid, **norepinephrine**, ± hydrocortisone	⭐ **IM adrenaline 0.5 mg (1:1000)**, fluids, antihistamine, steroid	Fluids, **norepinephrine/phenylephrine**, **atropine**

Table C: Cardiogenic Shock Profiles (Slide 22)

	Dry–cold	Wet–cold
Filling	Low (PCWP < 15)	High
Step 1	**Fluid challenge 250–500 mL**	**Inotrope** (dobutamine/milrinone)
Step 2	**Norepinephrine**	**Norepinephrine** if still shocked
Step 3	**Inotrope**	**Diuretic once SBP > 90**
Refractory	MCS/ECMO	Refractory AHF therapy / MCS

Table D: Obstructive Shock: Diagnosis and Fix

Cause	Key sign	Test	Fix
Tamponade	Beck's triad, pulsus paradoxus	**Echo** (RA/RV diastolic collapse)	**Pericardiocentesis**

Cause	Key sign	Test	Fix
Tension PTX	Absent breath sounds, tracheal shift	**Clinical** (X-ray/US)	**Needle → chest drain**
Massive PE	↑ JVP, hypoxaemia, RV strain	**CTPA / echo**	**Thrombolysis**
Aortic dissection	Tearing pain, pulse deficit	**CT angiography**	Surgery (type A)

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4. Rules, Exceptions, Patterns

- **Shock = tissue hypoperfusion**; hypotension is not required in pre-shock (compensated).
- **Pre-shock:** cold, clammy, ↑ CRT, **tachycardia**, oliguria. **Exceptions:** distributive (warm) and **neurogenic (bradycardia)**.
- **Cardiogenic = the only high-PCWP shock; distributive = the only low-SVR shock.**
- **Obstructive and cardiogenic both raise the JVP**; lungs are wet in cardiogenic, clear in obstructive.
- **MI is the commonest cause of cardiogenic shock → revascularise.**
- **Dry-cold: fluid only if PCWP < 15. Wet-cold: no fluid; inotrope first.**
- **Diuretics only once SBP > 90.**
- **Norepinephrine is the preferred vasopressor.**
- **Lactate** measures global hypoperfusion; trend it.
- **FAST** for haemorrhage; **tryptase** for anaphylaxis; **CTPA/V/Q** for PE.
- **Refractory shock → hydrocortisone, bicarbonate, MCS/ECMO.**

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5. Clinical Correlations

- **Post-MI day 3–5 new murmur + shock** → **papillary-muscle rupture (acute MR)** or **VSD** → urgent echo and surgery (CAD lecture).
- **β-blocker/CCB overdose** → cardiogenic shock with bradycardia → *[added]* glucagon (β-blocker), calcium, **high-dose insulin**, pacing.
- **Spinal/epidural anaesthesia or cord injury** → neurogenic shock (lecture case).
- **Trauma:** haemorrhagic shock (FAST, pelvic X-ray) vs tension PTX vs tamponade vs neurogenic shock.
- **Adrenal crisis** → refractory shock unresponsive to pressors → **IV hydrocortisone 100 mg** (Adrenal II).
- **Sepsis** (next ID lectures) → distributive shock; lactate > 2 + pressors = septic shock.
- **Massive PE** (Pulmonary Vascular lecture) → obstructive shock → thrombolysis.
- **Tamponade** after cardiac surgery or in malignancy (Pericardial lecture).
- **ECMO** for ARDS (VV) vs cardiogenic shock (VA) (ARDS lecture).

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6. Rapid Revision (Last-Day)

1. **Shock** = life-threatening circulatory failure → tissue hypoxia, microcirculatory disturbance → metabolic derangement → irreversible organ damage.
2. **Types:** hypovolaemic, **cardiogenic**, obstructive, distributive (**septic, anaphylactic, neurogenic**).
3. **Stages:**

- **pre-shock:** vasoconstriction (cold, clammy, ↑ CRT; not in distributive), tachycardia (not in neurogenic), oliguria, fluid shift into the vessels
- **progressive:** hypotension, tachypnoea, capillary pooling, microthrombi, **DIC**, acidosis, confusion
- **end-organ:** autonomic dysfunction, myocardial ischaemia, lysosomal enzymes, cytokines, **gut ischaemia → sepsis** → MOF

4. **Haemodynamics:**
   - hypovolaemic: CO ↓, PCWP ↓, SVR ↑
   - **cardiogenic: CO ↓↓, PCWP ↑, SVR ↑**
   - obstructive: CO ↓↓, PCWP ↓/↑, SVR ↑
   - **distributive: SVR ↓↓**, PCWP ↓, CO ↑ early (septic/anaphylactic) or ↓ (neurogenic/late)
5. **Routine tests: lactate**, ABG, BMP, renal, LFTs, coagulation.
6. **Haemorrhagic:** serial CBC (may be normal early), type and screen, **FAST**, X-ray (pelvis, haemothorax), DRE.
7. **Non-haemorrhagic hypovolaemic:** fever work-up, **CT abdomen** (pancreatitis, obstruction), GI losses, urine electrolytes.
8. **Cardiogenic tests:** BNP vs baseline, **ECG**, **troponin**, **echo** (valves, function, effusion).
9. **Obstructive tests:** echo/POCUS, CXR/CT (PTX), **CTPA/V/Q** (PE), CT (dissection).
10. **Distributive tests:** cultures + CRP/ESR/procalcitonin (septic); **tryptase** (anaphylaxis); CT/MRI spine/brain (neurogenic).
11. **Cardiogenic causes: MI (commonest)**, arrhythmia, HF, cardiomyopathy, myocarditis, **VSD/rupture**, **severe AR/MR**, blunt trauma, **β-blockers/CCBs**.
12. **Cardiogenic spiral:** ↓ CO → hypoperfusion → ↓ vasoreactivity → hypoxaemia → organ dysfunction → ↓ CO.
13. **Treatment: revascularise MI.** - **Dry–cold:** fluid only if hypotensive/PCWP < 15 (250–500 mL) → **norepinephrine** → inotrope (**dobutamine, milrinone, dopamine**). - **Wet–cold: inotrope** → norepinephrine → **diuretic once SBP > 90** → refractory AHF.
14. **Obstructive causes:** ↓ filling (**tamponade**, constriction, RCM), ↓ venous return (**tension PTX**, tumour), ↑ afterload (**massive PE**, dissection, AS, emboli, severe PH).
15. **Obstructive treatment: pericardiocentesis**; **thrombolysis**; **needle thoracostomy → chest tube**.
16. **Complications:** CV collapse, **AKI**, pulmonary oedema.
17. **Refractory shock: hydrocortisone, bicarbonate, MCS, ECMO** (severe ARDS, refractory respiratory or heart failure).

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7. Exam-Style Questions

Q1 (lecture case, slides 5–6). A 28-year-old has **intrathecal bupivacaine (L3–L4)** for a skin graft. 30 minutes later: dyspnoea, syncope, **HR 38**, **BP 82/50**, **flushed skin**, sensory level above the navel, clear lungs. Most likely mechanism? **A. Loss of sympathetic vascular tone** · B. Bacterial toxin vasodilation · C. PA thrombus · D. Volume depletion · E. Bone-marrow embolism · F. Type I hypersensitivity

Answer: A *[reasoning added; key not on slides]*.

- **High spinal block** → sympathectomy → vasodilatation and venous pooling; block of cardioaccelerator fibres (T1–T4) → **bradycardia**.
- **Hypotension + bradycardia + warm, flushed skin** = **neurogenic/distributive** physiology (slide 10: "neurogenic shock may present with bradycardia").
- Septic, anaphylactic, PE and hypovolaemic shock are all **tachycardic**.
- Fat/marrow embolism occurs within 72 h of a fracture, not at 4 weeks.

Q2. A 62-year-old 6 hours into an anterior STEMI has BP 78/50, HR 118, cold clammy peripheries, confusion, **bibasal crackles**, lactate 5.2. Echo: LVEF 20%, no mechanical complication. Best immediate strategy? A. 2 L crystalloid bolus · **B. Emergency PCI of the culprit vessel with inotrope/vasopressor support (norepinephrine ± dobutamine)** · C. IV furosemide 80 mg now · D. IV metoprolol · E. Nitroglycerin infusion

Answer: B (slides 20, 22).

- **MI is the commonest cause** → **revascularisation**.
- He is **wet–cold** → **inotrope**, ± norepinephrine.
- Why not the others:
- **Diuretics only once SBP > 90.**
- Fluids would worsen the oedema.
- **β-blockers and nitrates are contraindicated** in shock.

Q3. After a stab wound to the chest, a man has BP 70/40, HR 130, **distended neck veins**, **muffled heart sounds** and clear, equal breath sounds. Immediate treatment? A. Needle thoracostomy · **B. Pericardiocentesis** · C. Thrombolysis · D. Norepinephrine alone · E. 2 L crystalloid

Answer: B (slides 23–24).

- **Beck's triad** = **tamponade** → obstructive shock → **pericardiocentesis** (*[added]* definitive: thoracotomy).
- Tension PTX would show **absent breath sounds** and tracheal deviation → needle thoracostomy.

Q4. Which haemodynamic profile is typical of cardiogenic shock? A. CO ↑, PCWP ↓, SVR ↓ · B. CO ↓, PCWP ↓, SVR ↑ · **C. CO ↓↓, PCWP ↑, SVR ↑** · D. CO ↓, PCWP ↓, SVR ↓↓

Answer: C (slide 13).

- A = early septic shock.
- B = hypovolaemic.
- D = neurogenic/late distributive.

Q5. A patient remains hypotensive on escalating norepinephrine despite source control and adequate fluids for septic shock. Which rescue therapy does the lecture list? A. Stop vasopressors · **B. IV hydrocortisone** · C. Oral prednisolone next day · D. β-blocker · E. More crystalloid indefinitely

Answer: B (slide 26).

- **Corticosteroids (hydrocortisone)**, **bicarbonate**, **MCS/ECMO** are the rescue therapies for refractory shock.
- *[added]* Hydrocortisone 200 mg/day in septic shock on vasopressors.

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8. Exam Pitfalls & Traps

Trap	Correct thinking
Shock = low BP	❌ Shock = **tissue hypoperfusion**; BP may be maintained in pre-shock (check lactate, urine output, CRT)
All shock is tachycardic	❌ **Neurogenic** (and β-blocker/CCB toxicity) → **bradycardia**
All shock has cold peripheries	❌ **Distributive** (early septic, anaphylactic, neurogenic) = **warm**
Cardiogenic shock: give fluids	❌ Only if **dry–cold with PCWP < 15** (small challenge); wet–cold → **inotrope**, no fluids
Furosemide straight away in cardiogenic shock	❌ **Diuretics only once SBP > 90**
Dopamine is the first-line pressor	⚠️ **Norepinephrine** is preferred (*[added]* dopamine → more arrhythmias)
Nitrates / β-blockers in cardiogenic shock	❌ Worsen hypotension and the pump; avoid
Raised JVP = cardiogenic shock	⚠️ **Obstructive** shock also raises the JVP (tamponade, tension PTX, PE); check the lungs and echo
Waiting for a CXR in suspected tension PTX	❌ **Clinical** diagnosis; decompress first
Normal Hb excludes haemorrhagic shock	❌ **Hb can be normal initially**; repeat CBC
Anaphylaxis diagnosed by tryptase	❌ **Clinical**; tryptase only if uncertain; treat with **IM adrenaline** immediately
RCM as a cause of cardiogenic shock only	⚠️ The slide lists **restrictive CM and constriction** under **obstructive** (↓ diastolic filling)
IABP routinely in MI shock *[added]*	❌ No mortality benefit (IABP-SHOCK II); Impella/VA-ECMO in selected patients